Butterworth Filter

Finding Poles Step by Step

A beginner-friendly study note with diagrams and a fully worked example

Covers: pole formula · pole circle · left-half-plane stability · 6th-order example ($\omega_c = 1$ rad/s)

0 1 Contents

1 What is a Butterworth Filter? 1

2 Why Do We Need to Find Poles? 1

3 The Pole Formula — Deriving Step by Step 2
- 3.1 Step 1 — Write the denominator polynomial . . . 2
- 3.2 Step 2 — Solve for the roots . . . 2
- 3.3 Step 3 — What this means geometrically . . . 2

4 The Pole Circle — Visualised 3

5 Two Big "Why" Questions 3
- 5.1 Why do we use only the LEFT half of the circle? . . . 3
- 5.2 Why are there no poles at ± 90 (i.e., at $\pm j\omega_c$)? . . . 4

6 Step-by-Step Algorithm to Find Poles 4

7 Worked Example: 6th-Order Butterworth, $\omega_c = 1$ rad/s 5
- 7.1 Setup . . . 5
- 7.2 Calculate Each Pole Angle . . . 5
- 7.3 Convert to Cartesian $(\sigma + j\omega)$ Coordinates . . . 5
- 7.4 The Pole Diagram for 6th Order . . . 6
- 7.5 Form the Transfer Function . . . 6

8 Quick Reference Summary 7

1 1 What is a Butterworth Filter?

Quick Intuition

Imagine you want to pass low-frequency signals (like music bass) and block high-frequency noise. A Butterworth filter does exactly that, and it does it **as smoothly as possible** — no ripple in the passband, no spikes in the stopband. It is called a *maximally flat* filter.

The **magnitude response** of an N-th order Butterworth low-pass filter is:

Butterworth Magnitude Response

$$|H(j\omega)|^2 = \frac{1}{1 + \left(\dfrac{\omega}{\omega_c}\right)^{2N}}$$

- N = filter order (higher N = sharper roll-off)
- ω_c = cutoff (3 dB) angular frequency in rad/s
- At $\omega = \omega_c$: $|H| = \dfrac{1}{\sqrt{2}} \approx 0.707 \;\Rightarrow\; -3\,\text{dB}$ point

2 1 Why Do We Need to Find Poles?

To *implement* the filter (in hardware or software) we need a transfer function $H(s)$, not just its magnitude. The poles of $H(s)$ completely define the filter. Once we know the poles, we can write:

$$H(s) = \frac{\omega_c^N}{\prod_{k=1}^{N}(s - s_k)}$$

where s_k are the poles. **Finding the poles is therefore the central design step.**

3 1 The Pole Formula — Deriving Step by Step

3.1 Step 1 — Write the denominator polynomial

We start from $|H(j\omega)|^2$ and replace $j\omega \to s$ (analytic continuation):

$$H(s)\,H(-s) = \frac{1}{1 + \left(\dfrac{s}{j\,\omega_c}\right)^{2N}} = \frac{\omega_c^{2N}}{\omega_c^{2N} + (-1)^N s^{2N}}$$

The **poles** of $H(s)\,H(-s)$ are the $2N$ roots of:

$$\omega_c^{2N} + (-1)^N s^{2N} = 0$$

3.2 Step 2 — Solve for the roots

Rearrange:

$$s^{2N} = (-1)^{N+1}\,\omega_c^{2N}$$

Using Euler's formula, the $2N$ solutions (roots on a circle of radius ω_c) are:

Pole Formula

$$\boxed{s_k = \omega_c \exp\left[j\left(\frac{\pi}{2} + \frac{(2k-1)\pi}{2N}\right)\right], \qquad k = 1,\, 2,\, \ldots,\, 2N}$$

Equivalently in Cartesian form:

$$s_k = \omega_c \left[\cos\left(\frac{\pi}{2} + \frac{(2k-1)\pi}{2N}\right) + j\,\sin\left(\frac{\pi}{2} + \frac{(2k-1)\pi}{2N}\right)\right]$$

3.3 Step 3 — What this means geometrically

All $2N$ poles lie **equally spaced on a circle of radius** ω_c in the complex s-plane. They are separated by an angle of $\frac{\pi}{N}$ radians $= \frac{180}{N}$.

4 1 The Pole Circle — Visualised

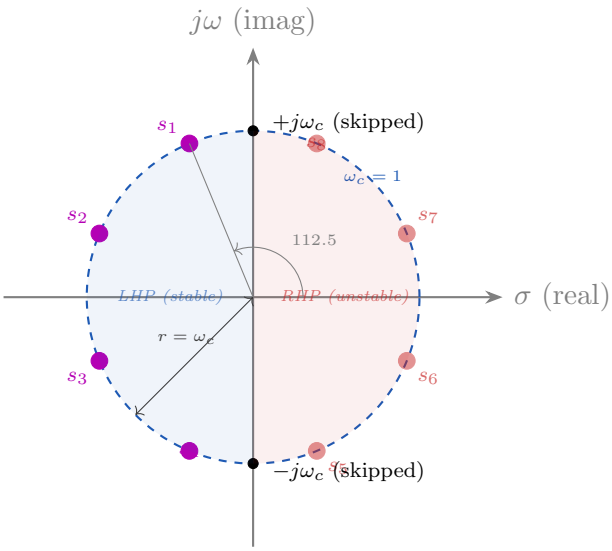

Figure 1: Pole circle for $N = 4$ (8 poles total). Purple = used (LHP). Red = discarded (RHP).

5 1 Two Big "Why" Questions

5.1 Why do we use only the LEFT half of the circle?

Stability Requirement!

A system with a pole at $s = \sigma + j\omega$ has a time-domain response that contains a term $e^{\sigma t}$.

- If $\sigma > 0$ (right half plane, RHP): $e^{\sigma t} \to \infty$ as $t \to \infty$ — **the system blows up!**
- If $\sigma < 0$ (left half plane, LHP): $e^{\sigma t} \to 0$ as $t \to \infty$ — **the system is stable.**
- If $\sigma = 0$ (imaginary axis): marginally stable (oscillates forever).

We need a stable filter, so we keep *only* the N poles in the left half plane. The right-half-plane poles belong to $H(-s)$, the "mirror" system — we throw them away.

5.2 Why are there no poles at ± 90 (i.e., at $\pm j\omega_c$)?

The Symmetry Argument

The pole angles are:

$$\theta_k = \frac{\pi}{2} + \frac{(2k-1)\pi}{2N}, \quad k = 1, 2, \ldots, 2N$$

The spacing between poles is $\frac{\pi}{N}$ (i.e., $\frac{180}{N}$), and the formula starts at $\frac{\pi}{2} + \frac{\pi}{2N}$, which is always *offset* from 90 by $\frac{\pi}{2N}$.

In plain English: The formula is designed so that poles never land exactly on the imaginary axis ($\pm j\omega_c$, i.e., ± 90). If they did, the filter would have poles right on the stability boundary — it would oscillate forever and be useless as a practical filter. The $(2k-1)$ factor in the formula guarantees an odd-multiple offset that always misses 0, ± 90, and 180 exactly.

Summary table for $N = 4$: Poles are at angles $112.5, 157.5, 202.5, 247.5$ (LHP) — never at 90 or 270.

6 1 Step-by-Step Algorithm to Find Poles

Given: Filter order N, cutoff frequency ω_c

Step 1. There will be $2N$ poles in total, all on a circle of radius ω_c.
Step 2. Compute the angle for each pole $k = 1, 2, \ldots, N$ (LHP only):

$$\theta_k = 90 + \frac{(2k-1) \times 180}{2N}$$

Step 3. Convert to Cartesian coordinates:

$$s_k = \omega_c \cos(\theta_k) + j\,\omega_c \sin(\theta_k)$$

Step 4. The N poles $s_1, s_2, \ldots, s_N$ are your Butterworth poles. (Discard anything with $\mathrm{Re}(s_k) > 0$.)
Step 5. Write the transfer function:

$$H(s) = \frac{\omega_c^N}{\prod_{k=1}^{N}(s - s_k)}$$

7 1 Worked Example: 6th-Order Butterworth, $\omega_c = 1$ rad/s

Problem Statement

Find all poles of a 6th-order Butterworth low-pass filter with cutoff frequency $\omega_c = 1$ rad/s.

7.1 Setup

$N = 6$, $\omega_c = 1$. There will be $2N = 12$ poles on the unit circle. We use only the $N = 6$ in the LHP.

Pole spacing $= \dfrac{180}{N} = \dfrac{180}{6} = 30$.

7.2 Calculate Each Pole Angle

$$\theta_k = 90 + \frac{(2k-1) \times 180}{2 \times 6} = 90 + (2k-1) \times 15$$

k	Formula	θ_k	Check (LHP?)
1	$90 + 1 \times 15$	**105**	✓ (LHP)
2	$90 + 3 \times 15$	**135**	✓ (LHP)
3	$90 + 5 \times 15$	**165**	✓ (LHP)
4	$90 + 7 \times 15$	**195**	✓ (LHP)
5	$90 + 9 \times 15$	**225**	✓ (LHP)
6	$90 + 11 \times 15$	**255**	✓ (LHP)

All 6 angles are between 90 and 270 — confirming they are all in the LHP.

7.3 Convert to Cartesian ($\sigma + j\omega$) Coordinates

Since $\omega_c = 1$, we just compute $\cos\theta_k$ and $\sin\theta_k$:

Pole	θ_k	$\sigma_k = \cos\theta_k$	$\omega_k = \sin\theta_k$	s_k
s_1	105	-0.2588	$+0.9659$	$-0.2588 + j\,0.9659$
s_2	135	-0.7071	$+0.7071$	$-0.7071 + j\,0.7071$
s_3	165	-0.9659	$+0.2588$	$-0.9659 + j\,0.2588$
s_4	195	-0.9659	-0.2588	$-0.9659 - j\,0.2588$
s_5	225	-0.7071	-0.7071	$-0.7071 - j\,0.7071$
s_6	255	-0.2588	-0.9659	$-0.2588 - j\,0.9659$

Key observation: Poles come in complex conjugate pairs: $s_1 \leftrightarrow s_6$, $s_2 \leftrightarrow s_5$, $s_3 \leftrightarrow s_4$.

7.4 The Pole Diagram for 6th Order

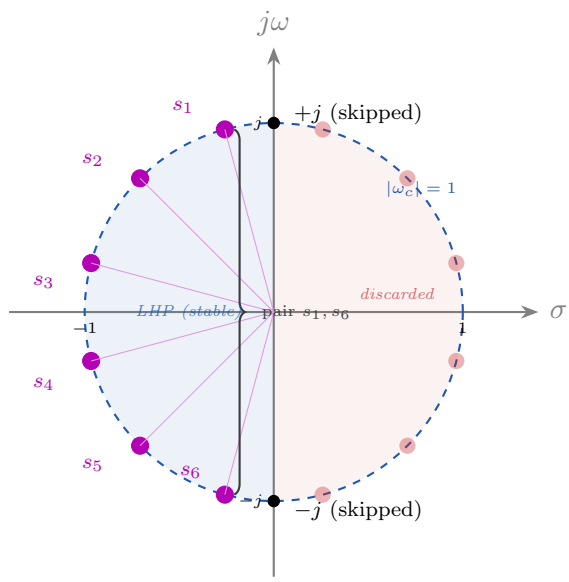

Figure 2: 6th-order Butterworth pole diagram ($\omega_c = 1$). Purple crosses = used poles. Faint red = discarded. Note $\pm j$ positions are always skipped.

7.5 Form the Transfer Function

Group conjugate pairs into real-coefficient quadratics using $(s - s_k)(s - s_k^*)$:

$$(s - s_k)(s - s_k^*) = s^2 - 2\sigma_k\, s + \omega_c^2$$

Since $\omega_c = 1$:

Quadratic Factors

$$\begin{aligned}
&\textbf{Pair 1:} \quad (s_1,\, s_6) \;\rightarrow\; s^2 + 0.5176\, s + 1 \\
&\textbf{Pair 2:} \quad (s_2,\, s_5) \;\rightarrow\; s^2 + 1.4142\, s + 1 \\
&\textbf{Pair 3:} \quad (s_3,\, s_4) \;\rightarrow\; s^2 + 1.9319\, s + 1
\end{aligned}$$

The $-2\sigma_k$ coefficients come from $-2\cos\theta_k$:

- Pair 1: $-2\cos 105 = -2(-0.2588) = 0.5176$
- Pair 2: $-2\cos 135 = -2(-0.7071) = 1.4142$
- Pair 3: $-2\cos 165 = -2(-0.9659) = 1.9319$

Therefore, the complete 6th-order transfer function is:

Final Transfer Function ($\omega_c = 1$ rad/s)

$$H(s) = \frac{1}{(s^2 + 0.5176\, s + 1)(s^2 + 1.4142\, s + 1)(s^2 + 1.9319\, s + 1)}$$

All three quadratic factors have positive coefficients — confirming stability (all poles in LHP).

8 1 Quick Reference Summary

Concept	Rule / Formula
How many poles to use?	N poles (one per filter order), out of $2N$ total
Where do poles lie?	On a circle of radius ω_c in the s-plane
Pole angle formula	$\theta_k = 90 + \dfrac{(2k-1)\times 180}{2N}, \quad k = 1 \ldots N$
Pole spacing	$\dfrac{180}{N}$ degrees between adjacent poles
Why left half only?	Right-half poles $\Rightarrow e^{\sigma t}$ grows $\Rightarrow$ unstable
Why skip ± 90?	Formula offset ensures poles never land on imaginary axis
Conjugate pairs	Poles come in $\sigma \pm j\omega$ pairs; combine into real quadratics

Final Tip for Beginners

The 3-sentence summary of Butterworth pole finding:

(1) Draw a circle of radius ω_c in the s-plane.

(2) Place N equally-spaced points on the *left half* of that circle, starting $\frac{90}{N}$ away from the top.

(3) Those points are your poles — plug them into $H(s)$ and you're done!